Node Modules

Agenda

1. Introduction to Node Modules
2. User Defined Modules
3. Core Modules
4. Third-party Modules

1. Introduction to Node Modules

- Module is a simple or complex functionality organized in single or multiple JavaScript files.
- It helps in reusability, maintainability, and modularity in Node.js applications.
- exports is a global object available to all the modules.
- Node modules can be imported into Node applications using the require() function
- There are 3 types of modules:
  1. Core modules
  2. Local modules
  3. User defined modules

1. User Defined Modules

- Created by the developer locally in the Node.js application
- Creating a User-defined module:
  - Create a separate .js file containing require functions, variables and objects. If any of this to be exposed outside the module, it should be stuffed into exports object like
```
exports.add = function (a, b) {
    return a+b;
};
```
- Using the User-Defined (Custom) Module:
  - var cm = require('./customModule.js'); where customModule.js is a separate file containing the functionality

3. Core Modules

- Set of built in modules of that are part of nodejs runtime installation are called core modules.
- The core modules include bare minimum functionalities of Node.js.
- Examples of core modules
  - http
  - url
  - fs etc
- In order to use Node.js core modules, it needs to be imported using require() function.

3. Core Modules

- Using http module
  - The http module is a core module that provides functionality for creating HTTP servers
  - It allows to handle incoming HTTP requests, send HTTP responses.
  - Node JS application when run starts a web server on the specified port
  - Clients can raise request using this port no and machin's IP address on the network.

3. Core Modules

- Using url module
  - It provides functions for URL parsing and get information about different parts of URL like hostname, pathname, query parameters etc
- Using fs modules
  - It provides file system-related functionality
  - It perform various operations such as reading from and writing to files, creating directories, renaming files, deleting files etc
  - It provides both asynchronous (non-blocking) and synchronous (blocking) functions where use of asynchronous is recommended

4. Third-party Modules

- These modules are created and maintained by NodeJS community.
- These modules can be installed via npm (Node Package Manager).
- Examples of third party modules are express, mongoose, etc.

4. Third-party Modules

- Node Package Manager(NPM)
  - Command line tool that installs, updates or uninstalls Node.js modules(packages) in the application.
  - NPM performs the operation in two modes: global and local.
  - Local installation: It is allowed to be used by the node applications only in the respective direcory in which it is installed
  - Global installation: It is allowed to be used by node applications anywhere in the fie system
  - NPM can even update the already installed packages
  - NPM can even uninstall packages if required